AP CALCULUS AB

Unit 1 • Worksheet 1-9A • Answer key

Worksheet 1-9A • IVT, EVT, and MVT

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

A runner's distance $D(t)$ is differentiable. "Was the runner ever going exactly as fast as her average speed?" Which theorem answers this?

(A) IVT
(B) EVT
(C) MVT
(D) None

Solution

Answer: (C). A speed (slope) equal to an average rate: MVT.

Question 2

NO CALCULATOR

The height $h(t)$ of a kite is continuous on $[0, 30]$. "Was the kite ever exactly 25 m high?" given $h(0) = 2$ and $h(30) = 40$. Which theorem answers this?

(A) IVT
(B) EVT
(C) MVT
(D) None

Solution

Answer: (A). A height between $h(0)$ and $h(30)$: IVT.

Question 3

NO CALCULATOR

"Was there a highest point of the kite on $[0, 30]$?" Which theorem answers this?

(A) IVT
(B) MVT
(C) None
(D) EVT

Solution

Answer: (D). Continuous on a closed interval: an absolute maximum exists (EVT).

Question 4

NO CALCULATOR

Which theorem needs a **closed** interval and gives an absolute maximum and minimum?

(A) IVT
(B) EVT
(C) MVT
(D) All three

Solution
Answer: (B). The EVT.

Questions 5 to 7 refer to the table. f is differentiable.

x	0	3	5	8
$f(x)$	2	-4	6	6

Question 5

NO CALCULATOR

Which statement is guaranteed?

(A) $f(c) = 0$ for some c in $(5, 8)$
(B) $f(c) = 0$ for some c in $(0, 3)$
(C) $f'(c) = 0$ for some c in $(0, 3)$
(D) $f'(c) = -2$ for some c in $(3, 5)$

Solution
Answer: (B). IVT: 0 is between 2 and -4.

Question 6

NO CALCULATOR

The MVT guarantees a c in $(3, 5)$ with $f'(c) =$

(A) 1
(B) -5
(C) 5
(D) 10

Solution
Answer: (C). $\frac{6-(-4)}{2} = 5$.

Question 7

NO CALCULATOR

Must there be a c in $(5, 8)$ with $f'(c) = 0$?

(A) Yes, by the MVT.
(B) Yes, by the IVT.
(C) No, f is constant there.
(D) No, the MVT needs $f(a) \neq f(b)$.

Solution
Answer: (A). $\frac{6-6}{3} = 0$ and f is differentiable.

Question 8

NO CALCULATOR

h is continuous on $[0, 4]$ with a sharp corner at $x = 2$. Which theorems apply on $[0, 4]$?

(A) All three
(B) MVT only
(C) IVT and EVT only
(D) None

Solution
Answer: (C). The MVT also needs differentiability.

Question 9 (Free response)

NO CALCULATOR

g is differentiable.

x	1	2	4	7
$g(x)$	6	1	5	11

(a) Must $g(c) = 3$ for some c in $(1, 2)$? Justify.
(b) Must $g'(c) = 2$ for some c in $(2, 4)$? Justify.
(c) Must g have an absolute minimum on $[1, 7]$? Justify.
(d) $k(x) = \frac{1}{x-3}$ has $k(2) = -1$ and $k(4) = 1$. Does the IVT guarantee $k(c) = 0$ in $(2, 4)$? Explain.

Solution
(a) Yes, IVT: g is continuous (differentiable) and $1 < 3 < 6$.
(b) Yes, MVT: $\frac{5-1}{4-2} = 2$.
(c) Yes, EVT: continuous on the closed interval $[1, 7]$; the minimum is at most 1.
(d) No: k is not continuous at $x = 3$ (vertical asymptote), so the IVT does not apply.